Vungu Rural District Council

Nextcloud and LibreOffice in the Vungu Planning Portal

What they would be used for, the benefits and drawbacks, how to integrate them, and what it would cost.

Document	Technical proposal and cost estimate
Date	26 September 2026
Scope	vunguwebapp-frontend (Vue 3) and vunguwebapp-backend (Fastify + PostGIS)
Status	For decision. No code has been changed.

Prices are estimates from public list prices found in September 2026 and must be confirmed with each vendor before a budget is approved. Euro amounts are converted at about 1 EUR = 1.10 USD.

1. Summary

Yes, both can be used in the portal, and one of them fixes a problem the portal already has.

Today, files uploaded to the portal (citizen documents, inspection photos, plan-review files and application attachments) are written to the backend server's local disk. The production backend runs on Render, and **render.yaml** does not attach a persistent disk. Render wipes that disk on every redeploy, so **uploaded files are lost each time the backend is redeployed**. The team already found this problem for decision letters (migration 087 moved those into the database). The other uploads still have it. In addition, server.js serves the *uploads/* folder publicly at */uploads/* with no login or role check. This includes citizen ID documents. The file names are random, but anyone who has a link can download the file.

The proposal is to run **Nextcloud** as the council's document store and **Collabora Online** (LibreOffice running in the browser, sold with Nextcloud as *Nextcloud Office*) as the document editor. The portal database remains the system of record for applications, decisions and audit. Nextcloud stores the files. LibreOffice edits Word documents and converts them to PDF.

> **Recommendation.** Self-host Nextcloud Community Edition with a paid Collabora Business subscription on one server in a South African cloud region. This is Option A in Section 8. The estimated running cost is **USD 320 to 370 per month**, which is inside the approved hosting budget. The integration needs about **16 to 22 developer-days** (4 to 5 weeks), done in three phases. Phase 1 stops uploads being lost and should go ahead even if in-browser editing is postponed.

Key figures

Item	Estimate
Monthly running cost, recommended option	USD 320 to 370
Monthly running cost, with full vendor support (Enterprise)	USD 870 to 960
One-time integration effort	16 to 22 developer-days, plus 5 to 7 for optional Phase 3
Staff training	Half a day per role group
Ongoing administration	About 1 day per month (updates, backup checks)

2. What each product is

Product	What it is	Licence
Nextcloud Hub	Self-hosted file server, similar to Google Drive or SharePoint. It provides folders, version history, sharing links, a recycle bin, desktop and mobile sync, and an API (WebDAV and OCS) that other systems can use.	AGPL-3.0, free. Paid Enterprise support is optional.
LibreOffice	Open-source office suite (Writer, Calc, Impress) that reads and writes .docx, .xlsx, .pptx and OpenDocument formats, and exports to PDF.	MPL-2.0, free
Collabora Online (Nextcloud Office)	The LibreOffice engine running on a server, with the editor shown in the browser. Several people can edit the same document at once. It plugs into Nextcloud through the *richdocuments* app and the WOPI protocol.	CODE edition is free but intended for testing and home use. Business subscription is paid.

In this document, **LibreOffice** means Collabora Online, because a web portal needs the browser version. The desktop LibreOffice application is only relevant if staff also want to edit files on their own computers.

3. How documents are handled in the portal today

This review looked at both repositories. The relevant findings are:

Area	Where	Current behaviour
Citizen ID and residency documents	backend src/routes/documents.js	Saved to *uploads/citizen-documents/<userId>/* on local disk. Checked with SHA-256, file signature and a malware scan.
Inspection photos, plan review files, application attachments	src/routes/inspections.js, plan-review.js, development-management.js	Saved to local disk under *uploads/* and served by server.js at */uploads*.
Decision letters	migration 087, generated_document.content	Stored as HTML in the database because of the disk problem. They cannot be edited as documents.
Generated PDFs	councilMapPdf, siteAssessmentPdf, wardProfilePdf, evidenceCertificatePdf	Built in code with pdfkit and pdf-lib. The layout is fixed, so staff cannot edit the wording after generation.
Word output	backend package docx 9.x	The library is installed, so .docx files can already be generated.
Letterhead, permits, planner documents	frontend src/documents/, views/planner/permit/	Rendered in the browser with jsPDF and pdf-lib.
Hosting	render.yaml	Render web service with no disk defined. The filesystem is ephemeral.

Conclusion. The portal can already produce documents. What it lacks is a) a durable place to keep files and b) a way for officers to edit the wording of a committee report, letter or permit before it is issued. Nextcloud provides (a) and LibreOffice provides (b).

4. What they would be used for

Nextcloud: the council's document store

- **Durable storage for every upload.** Citizen documents, inspection photos, survey plans, building plans and specialist referral responses would be saved to Nextcloud and would no longer be lost on redeploy.
- **One folder per application.** For example, */Applications/VRDC-2026-0142/{submission, referrals, inspections, reports, decisions}*, so every file for a case can be found in one place.
- **Version history.** Every save of a committee report is kept, so officers can see what changed between the draft and the version the committee approved.
- **Sharing with outside parties.** Password-protected, expiring links can be sent to consultants, the Ministry of Local Government, Environmental Management Agency (EMA) reviewers or surveyors, without giving them a portal account.
- **Staff file workspace (optional).** Staff can use the desktop or mobile sync client for non-case documents such as council minutes, templates, the local development plan and bylaws.
- **Records management.** Retention rules, a recycle bin and file-level audit support a paperless register.

LibreOffice (Collabora Online): editing and conversion

- **Committee reports.** The portal generates a first draft .docx from case data (useReportBuilder and Step 3). The planner then edits it in the browser in a Word-like editor before it goes to committee.
- **Decision letters, permits, conditions and enforcement notices.** The portal fills the template, the officer adjusts the wording, and the approved version is locked.
- **Conversion to PDF on the server.** The final .docx is converted to PDF/A for issuing and archiving. Citizens receive a PDF that cannot be edited.
- **Council templates.** The letterhead and standard letters are kept as .docx or .odt templates in Nextcloud. Administrative staff can change the wording without a developer or a code release.
- **Spreadsheets.** Fee schedules and register exports can be opened in the browser instead of being downloaded.

Who uses what

Role	Nextcloud	LibreOffice editor
Citizen	Never logs in to Nextcloud. Uploads and downloads go through the portal as they do today.	No
Planning clerk	Case folders, scanning intake documents	Acknowledgement letters, registers
Planner / EO planner	Case folders, sharing with referral agencies	Committee reports, decision letters, conditions
Inspector	Photos and evidence, including from mobile	Inspection reports, notices
EHO / Environment officer	Licensing evidence	Licences, compliance letters
Admin	Users, quotas, retention, backups	Template maintenance

5. Advantages

- **It fixes the lost-uploads problem.** This is the largest benefit, and it is needed with or without the editor.
- **Data stays under council control.** Both products run on servers the council chooses, which supports obligations under the Cyber and Data Protection Act [Chapter 12:07]. Legal advice should confirm the hosting location.
- **No per-seat Microsoft or Google licences.** The software is open source, and paid support is optional.
- **It fits the portal's self-hosting rule.** The frontend must not load third-party CDNs. Both products run on the council's own domain, unlike Google Docs or Microsoft 365.
- **Word documents are the output officers expect.** Staff edit .docx files, and the council issues PDFs.
- **Several people can edit at once.** A planner and a senior planner can review the same report together.
- **Version history and audit trail** for each document, in addition to the portal's own audit log.
- **Standard protocols** (WebDAV, WOPI, OCS REST). The backend does not need an SDK, only HTTP calls with Node's built-in fetch.
- **Templates move out of code.** Changing the wording of a letter becomes an administrative task rather than a code release.
- **It scales with the council.** Nextcloud can grow from 20 to several thousand users on the same design.

6. Disadvantages and risks

Disadvantage or risk	Impact	Mitigation
Another system to operate (PHP, Redis, database, cron, Collabora)	Medium	Use **Nextcloud All-in-One (AIO)**, a single Docker image that includes Collabora, Redis, the database and BorgBackup. Updates are applied with one click.
Frequent releases (about 3 major versions a year) and security patches	Medium	Schedule a monthly patch window and subscribe to Nextcloud security advisories.
Collabora uses a lot of memory (roughly 50 to 150 MB per open document)	Low for about 40 staff	Use 16 GB of RAM and monitor usage.
Complex Microsoft Word layouts do not always look identical	Low to medium	Build templates in LibreOffice and issue a PDF, not the .docx.
The free CODE edition is not supported for production and shows a notice above small usage limits	Medium	Buy the Collabora Business subscription (Option A).
Staff may need a second login	Medium	Phase 2 uses Nextcloud's Direct Editing API, so the portal opens the editor without a Nextcloud login. Single sign-on (OIDC) is optional in Phase 3.
Two copies of the truth	High if not managed	The portal database remains the system of record. Nextcloud only holds the file bytes, and the database stores the file id and path.
The editor must be allowed to load in a frame on the portal domain	Low	Start by opening the editor in a new tab. Add the portal to the Collabora and Nextcloud frame-ancestors setting if embedding is wanted later.
Latency from Zimbabwe to the server	Low to medium	Host in Johannesburg (Azure South Africa North) or Cape Town (AWS af-south-1), not in Europe or the US. Test latency before choosing.
One more server that could go down or be attacked	Medium	Daily off-site backups, a tested restore, 2FA for administrators, the ClamAV antivirus app, and a firewall that allows only 443.

7. How to integrate

7.1 Target architecture

Component	Role
Vue frontend (unchanged host)	Upload and download through the portal API as today. Adds an *Edit document* button for staff.
Fastify backend (Render)	Remains the only system citizens talk to. Checks roles, then reads and writes files in Nextcloud with a service account. Stores file ids in PostgreSQL.
Nextcloud AIO (new VM)	Stores files, version history and share links. It is reachable by staff and by the backend. Citizens never reach it directly.
Collabora Online (inside AIO)	Browser editor and the *convert-to* PDF service.

Component	Role
PostgreSQL (unchanged)	System of record: applications, decisions, audit, and the new file references.

Request flow for a citizen upload: browser, then the portal API (auth, signature and malware checks as today), then a WebDAV PUT to Nextcloud, then a database row that stores the Nextcloud path. For a download, the portal checks the role, streams the file from Nextcloud and returns it. Nextcloud URLs are never exposed to citizens.

7.2 Phase 0: stand up the server (about 3 to 4 days)

- Create a 4 vCPU / 16 GB VM in a South African region, with a separate data disk of about 500 GB.
- Install Nextcloud AIO and enable the Nextcloud Office (Collabora) and ClamAV options. Use the subdomain *docs.<council-domain>* with a Let's Encrypt certificate.
- Create groups that match the portal roles (planners, clerks, inspectors, eho, admin) and a *vungu-service* account with an app password.
- Configure BorgBackup to off-site storage and **test a restore** before Phase 1 ends.

7.3 Phase 1: move uploads into Nextcloud (about 6 to 8 days)

Add one small module to the backend, *src/services/fileStore.js*, and route the four routes that write to disk through it. If NEXTCLOUD_URL is not set, for example in local development, it continues to use the local disk.

```
// src/services/fileStore.js  (sketch; uses Node's built-in fetch, no new dependency)
const BASE = process.env.NEXTCLOUD_URL              // https://docs.example.gov.zw
const USER = process.env.NEXTCLOUD_USER             // vungu-service
const AUTH = 'Basic ' + Buffer.from(`${USER}:${process.env.NEXTCLOUD_APP_PASSWORD}`).toString('base64')
const dav  = (p) => `${BASE}/remote.php/dav/files/${USER}/${p.split('/').map(encodeURIComponent).join('/')}`

async function mkdirs(dir) {                        // MKCOL each level; 405 = already exists
  let acc = ''
  for (const part of dir.split('/').filter(Boolean)) {
    acc += '/' + part
    const r = await fetch(dav(acc), { method: 'MKCOL', headers: { Authorization: AUTH } })
    if (!r.ok && r.status !== 405) throw new Error(`MKCOL ${acc} ${r.status}`)
  }
}

async function put(path, buffer, mime) {
  await mkdirs(path.slice(0, path.lastIndexOf('/')))
  const r = await fetch(dav(path), { method: 'PUT', body: buffer,
    headers: { Authorization: AUTH, 'Content-Type': mime } })
  if (!r.ok) throw new Error(`PUT ${path} ${r.status}`)
}

async function get(path) {                          // returns a web stream to pipe to the reply
  const r = await fetch(dav(path), { headers: { Authorization: AUTH } })
  if (!r.ok) throw new Error(`GET ${path} ${r.status}`)
  return r
}

module.exports = { put, get }
```

- In *documents.js*, replace *fs.writeFile(diskPath, file.buffer)* with *fileStore.put(...)*. Keep the SHA-256, signature and malware checks before the upload.
- Replace the static */uploads* mount in server.js with a download route that checks the role and streams from *fileStore.get*. This also fixes the current situation where anyone who knows an upload URL can download the file.

- Write a one-off migration script that copies anything still in *uploads/* into Nextcloud and rewrites *storage_url*.
- Add a migration (next free number) that adds *nextcloud_path* to the tables that store *storage_url*.
- Add Render environment variables: NEXTCLOUD_URL, NEXTCLOUD_USER, NEXTCLOUD_APP_PASSWORD (all set to *sync: false*).

7.4 Phase 2: edit documents in the browser (about 7 to 10 days)

- **Generate:** the backend builds the draft .docx with the *docx* library it already has, or fills a council template stored in Nextcloud, and saves it to the case folder.
- **Open:** the backend requests a one-time editor URL from Nextcloud's Direct Editing API and returns it to the staff member, so no Nextcloud login is needed. The portal opens it in a new tab.
- **Finalise:** when the officer clicks *Issue*, the backend converts the .docx to PDF through Collabora, stores the PDF, marks the .docx read-only and records the decision in the portal's audit log with the officer's name.

```
# open an editor session (backend, as the service account)
POST {NEXTCLOUD_URL}/ocs/v2.php/apps/files/api/v1/directEditing/open
Headers: OCS-APIRequest: true, Authorization: Basic ..., Accept: application/json
Body:    path=/Applications/VRDC-2026-0142/reports/committee-report.docx
Returns: ocs.data.url   -> one-time editor link for the officer

# convert the final version to PDF (Collabora)
POST {COLLABORA_URL}/cool/convert-to/pdf      multipart form field: data=<docx bytes>
Returns: the PDF bytes -> fileStore.put('.../decisions/decision-letter.pdf', ...)
```

Note: edits made through a service-account session appear under that account in Nextcloud. The portal's own audit log therefore records who opened and who issued each document. If Nextcloud must show individual names, add single sign-on (Phase 3). Confirm the exact API behaviour against the Nextcloud version you install.

Frontend changes are small. Add an *Edit document* button to the relevant planner, inspector, EHO and clerk sections that calls the new endpoint and opens the returned URL. Show a loading state and an error state (for example *Document server unavailable*). Nothing new is loaded from a CDN.

7.5 Phase 3 (optional): staff workspace and single sign-on (about 5 to 7 days)

- Staff use Nextcloud directly for non-case documents, with desktop and mobile sync.
- Single sign-on: connect Nextcloud's *user_oidc* app to an identity provider such as Keycloak, or make the portal backend an OIDC provider, so each officer has one login and Nextcloud history shows individual names.
- Share links for referral agencies and consultants, created from the portal case page.

7.6 Security checklist

- Citizens never receive a Nextcloud URL. Every download goes through the portal's role check.
- Keep the existing checks (malware scan, file signature, size limit) **before** the file reaches Nextcloud.
- Store the service-account app password only in Render environment variables. Rotate it every quarter.
- Require 2FA for Nextcloud admins, apply Nextcloud's brute-force protection and allow only port 443 through the firewall.
- Allow Collabora to accept WOPI requests only from the Nextcloud host, and *convert-to* requests only from the backend's IP.
- Take daily encrypted off-site backups, keep them for 30 days and test a restore every quarter.

7.7 Effort

Phase	Work	Days
0	Server, AIO, TLS, groups, backups and restore test	3 to 4
1	fileStore module, 4 routes, protected downloads, migration, copying existing files, tests	6 to 8
2	DOCX generation and templates, direct editing, PDF finalisation, UI buttons, tests	7 to 10
3 (optional)	SSO, staff workspace, share links from the case page	5 to 7
Total	**Phases 0 to 2 (Phase 3 optional)**	**16 to 22 (21 to 29 with Phase 3)**

8. Costs

Assumptions: about **40 council staff users**. Citizens do not count because they never log in to Nextcloud. Storage starts at 500 GB. Hosting is in a South African cloud region for low latency from Zimbabwe. All figures are estimates and must be confirmed with each vendor.

8.1 Monthly running costs (USD)

Line item	A. Recommended: Community + Collabora Business	B. Full Enterprise support	C. Free software only
VM, 4 vCPU / 16 GB, South African region	170 to 200	170 to 200	170 to 200
Data disk, 500 GB SSD	45 to 60	45 to 60	45 to 60
Off-site backup storage (about 1 TB)	25 to 30	25 to 30	25 to 30
Nextcloud licence	0 (Community)	about 630 (Standard, minimum 100 users x EUR 68.94 per year)	0
Collabora / Nextcloud Office	about 76 (40 users x EUR 20.80 per year, Business)	Confirm in quote; often bundled as Nextcloud Office	0 (CODE, not supported for production)
TLS certificate / subdomain	0 (Let's Encrypt)	0	0
Total per month	**about 320 to 370**	**about 870 to 960**	**about 240 to 290**

8.2 One-time costs

Item	Estimate
Integration, Phases 0 to 2	16 to 22 developer-days
Optional Phase 3 (SSO, workspace)	5 to 7 developer-days
Converting council letters to LibreOffice templates	2 to 3 days of administrative and planning staff time
Staff training	Half a day per role group (about 5 sessions)

8.3 Comparison with the current budget

Option A adds about USD 320 to 370 per month and fits inside the approved hosting budget of USD 1,000 or more per month. Option B is only worth it if the council needs a vendor support contract with an SLA. Nextcloud Enterprise has a 100-user minimum, so a council with about 40 users would pay for 60 users it does not have. Option C costs the least, but Collabora does not support the free CODE edition in production. It is suitable for a pilot or a test server only.

The cheapest way to stop losing uploads, without an editor, is to attach a Render persistent disk, which costs about USD 0.25 per GB per month. That fixes Phase 1 only. It also ties the backend to a single instance and does not give version history, sharing or editing, so it is a stop-gap rather than a replacement for this proposal.

9. Alternatives considered

Alternative	Why it was not recommended
Object storage only (S3 or Cloudflare R2) plus the current PDF generators	Cheapest and simplest way to store files, but officers still cannot edit reports or letters, and there is no version history or sharing for staff. It is a reasonable choice if editing is not wanted.
ONLYOFFICE Docs instead of Collabora	Word compatibility is slightly closer. The free edition limits concurrent connections, and the paid edition costs more per server. Collabora fits better with Nextcloud AIO.
Microsoft 365 / SharePoint	Per-user licences, data held abroad, and editor scripts loaded from Microsoft's CDN, which breaks the portal's rule against third-party CDNs.
Google Workspace	Same concerns as Microsoft 365, plus weaker handling of .docx files.
Build an editor in the portal	Months of work to reproduce what LibreOffice already does.

10. Decisions needed and next steps

- **Approve Phase 1 now.** Uploads are being lost on every redeploy, whether or not Nextcloud is chosen.
- **Choose where to host** (Azure Johannesburg, AWS Cape Town or a local Zimbabwean data centre) after a one-day latency test from Vungu offices.
- **Choose a cost option:** A (recommended), B (with an SLA) or C (pilot only).
- **Name a template owner** in the planning department to convert current letters into LibreOffice templates.
- **Confirm legal requirements** for data residency and records retention under the Cyber and Data Protection Act and national archives rules.
- After approval, write a design spec and implementation plan in *docs/superpowers/*, as for earlier features.

Sources

- Nextcloud Enterprise tiers and 100-user minimum: opsily.com/hosting/nextcloud/nextcloud-enterprise-pricing; capterra.com/p/161572/NextCloud/pricing
- Collabora Online Business pricing (EUR 20.80 per user per year, 20 to 99 users): collaboraonline.com/subscriptions; capterra.com/p/10011676/Collabora-Online/pricing
- Nextcloud Office pricing discussion: help.nextcloud.com/t/nextcloud-office-collabora-online-pricing-for-20-users/206845
- VM reference pricing (4 vCPU / 16 GB from about USD 140 per month; African regions cost more): cloudprice.net/vm/Standard_D4s_v5; cloudprice.net/aws/ec2/instances/m6i.xlarge

- Codebase: vunguwebapp-backend render.yaml, server.js, src/routes/documents.js, migrations/087_generated_document_content.sql